AFFINE INTERSECTION CHARACTERIZATION OF SEPARATED MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4394-4429, following the statement at 4373-4392.

Allowed standard prerequisites (editorial).

- Open fibre products of affines. lemma-open-fibre-product (same chapter, cited at ~4399). Identifies U x_S V = U x_W V = Spec(A tensor_R B) as an affine open.
- Closed subscheme structure on preimage of closed immersion. lemma-closed-subspace-scheme (same chapter, cited at ~4402). Writes Delta^{-1}(U x_S V) -> U x_S V as Spec of a quotient ring.
- Affine open covering of the fibre product. lemma-affine-covering-fibre-product (same chapter, cited at ~4414). The opens U x_S V cover X x_S X.
- Closed immersion checked locally on the target. lemma-closed-local-target (same chapter, cited at ~4417). Reduces closedness of the diagonal to each piece.

Source reference. Original lemma 01JS: sources/stacks-schemes/pinned-original-source.tex, source lines 3448–3448. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HL: sources/stacks-schemes/pinned-original-source.tex, source lines 574–574. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01IN: sources/stacks-schemes/pinned-original-source.tex, source lines 1961–1961. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01JR: sources/stacks-schemes/pinned-original-source.tex, source lines 3415–3415. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes.*

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *If f is separated then for every pair of affine opens (U, V) of X which map into a common affine open of S we have*
   (a) *the intersection $U \cap V$ is affine.*
   (b) *the ring map $\mathcal{O}_X(U) \otimes_{\mathbf{Z}} \mathcal{O}_X(V) \to \mathcal{O}_X(U \cap V)$ is surjective.*

(2) *If any pair of points $x_1, x_2 \in X$ lying over a common point $s \in S$ are contained in affine opens $x_1 \in U$, $x_2 \in V$ which map into a common affine open of S such that (a), (b) hold, then f is separated.*

Proof. Assume f separated. Suppose (U, V) is a pair as in (1). Let $W = \mathrm{Spec}(R)$ be an affine open of S containing both $f(U)$ and $f(V)$. Write $U = \mathrm{Spec}(A)$ and $V = \mathrm{Spec}(B)$ for R-algebras A and B. By Lemma 01JR we see that $U \times_S V = U \times_W V = \mathrm{Spec}(A \otimes_R B)$ is an affine open of $X \times_S X$. Hence, by Lemma 01IN we see that $\Delta^{-1}(U \times_S V) \to U \times_S V$ can be identified with $\mathrm{Spec}((A \otimes_R B)/J) \to \mathrm{Spec}(A \otimes_R B)$ for some ideal $J \subset A \otimes_R B$. Thus $U \cap V = \Delta^{-1}(U \times_S V)$ is affine. Assertion (1)(b) holds because $A \otimes_{\mathbf{Z}} B \to (A \otimes_R B)/J$ is surjective.

Assume the hypothesis formulated in (2) holds. Clearly the collection of affine opens $U \times_S V$ for pairs (U, V) as in (2) form an affine open covering of $X \times_S X$ (see e.g. Lemma 01JS). Hence it suffices to show that each morphism $U \cap V = \Delta_{X/S}^{-1}(U \times_S V) \to U \times_S V$ is a closed immersion, see Lemma 01HL. By assumption (a) we have $U \cap V = \mathrm{Spec}(C)$ for some ring C. After choosing an affine open $W = \mathrm{Spec}(R)$ of S into which both U and V map and writing $U = \mathrm{Spec}(A)$, $V = \mathrm{Spec}(B)$ we see that the assumption (b) means that the composition

$$A \otimes_{\mathbf{Z}} B \to A \otimes_R B \to C$$

is surjective. Hence $A \otimes_R B \to C$ is surjective and we conclude that $\mathrm{Spec}(C) \to \mathrm{Spec}(A \otimes_R B)$ is a closed immersion. $\square$